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Exp No.- 5

Aim:- Simulation of automatic rain detector and wardrobe shelter using circuit design tinker card

Theory:-This setup involves an Arduino Uno connected to a soil moisture sensor and a servo motor. The circuit is likely designed to detect soil moisture levels and trigger an action (such as activating a watering system) when the soil becomes dry. When the soil moisture level drops below a certain threshold, the Arduino processes the signal and controls the servo motor to simulate a watering mechanism.

Components Required:

- Arduino Uno
- Soil Moisture Sensor (used to detect rain)
- Servo Motor (to move the shelter)
- Connecting Wires
- Power Supply

Connections:

- Soil Moisture Sensor:
  - VCC → Arduino 5V
  - GND → Arduino GND
  - Signal (A0) → Arduino Analog Pin A0
- Servo Motor:
  - VCC → Arduino 5V
  - GND → Arduino GND
  - Signal → Arduino Digital PWM Pin (e.g., D9)

Code Implementation:

- Read the soil moisture sensor values via Analog Input (A0).
- Set a threshold to detect rain.
- If rain is detected (high moisture value), trigger the servo motor to extend the shelter.
- If no rain is detected, retract the shelter.

Execution:

- Upload the code to the Arduino.
- Monitor the rain detection in real-time.
- When rain starts, the shelter covers the clothes.
- When the rain stops, the shelter retracts automatically.

rduino Code:-

Here’s an example code that reads the rain sensor and activates the motor (via relay) when rain is detected.

cpp
Copy

```cpp
int rainSensorPin = 2;     // Pin connected to the rain
sensor
int relayPin = 3;          // Pin connected to the
relay
int rainStatus = 0;        // Variable to store the
rain sensor value

void setup() {
  pinMode(rainSensorPin, INPUT);  // Set rain sensor
pin as input
  pinMode(relayPin, OUTPUT);      // Set relay pin as
output
```

```
  Serial.begin(9600);             // Start serial
communication for debugging
}

void loop() {
  rainStatus = digitalRead(rainSensorPin); // Read the
rain sensor
  Serial.println(rainStatus);             // Print
rain status to serial monitor

  if (rainStatus == HIGH) {  // Rain detected
    digitalWrite(relayPin, HIGH);  // Activate relay
(close shelter)
    Serial.println("Rain detected. Sheltering...");
  } else {
    digitalWrite(relayPin, LOW);  // Deactivate relay
(open shelter)
    Serial.println("No rain. Shelter open.");
  }

  delay(1000);  // Wait for 1 second before checking
again
}
```

Screenshots:-

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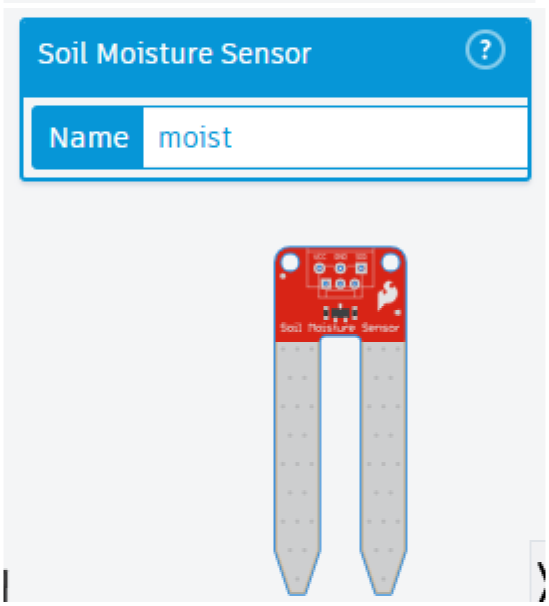

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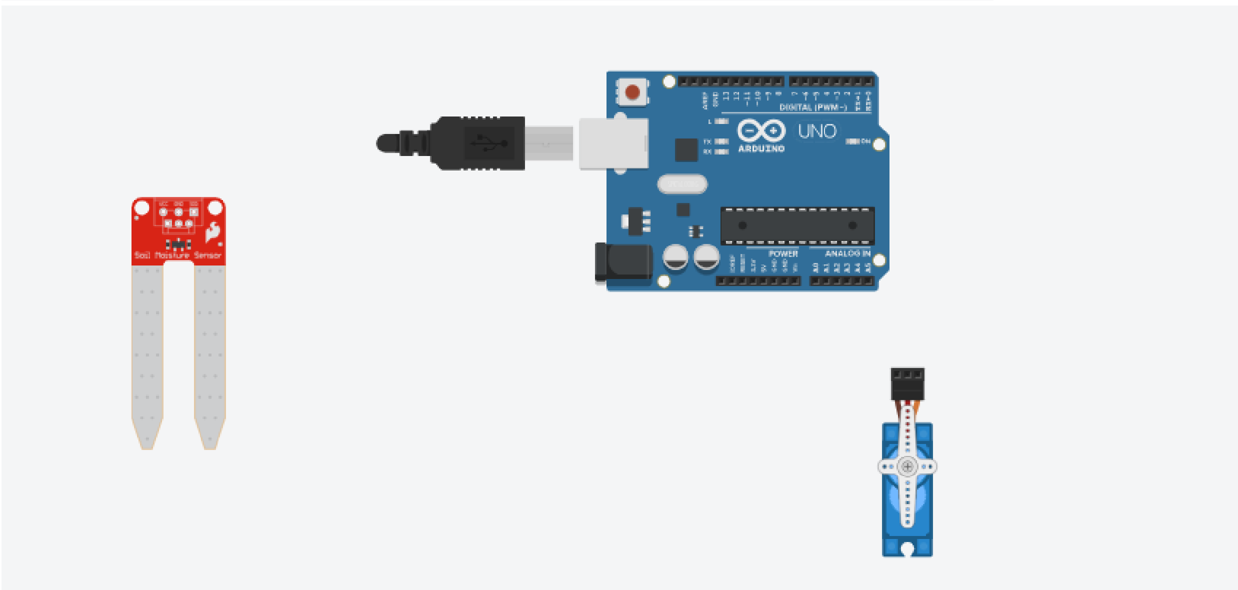

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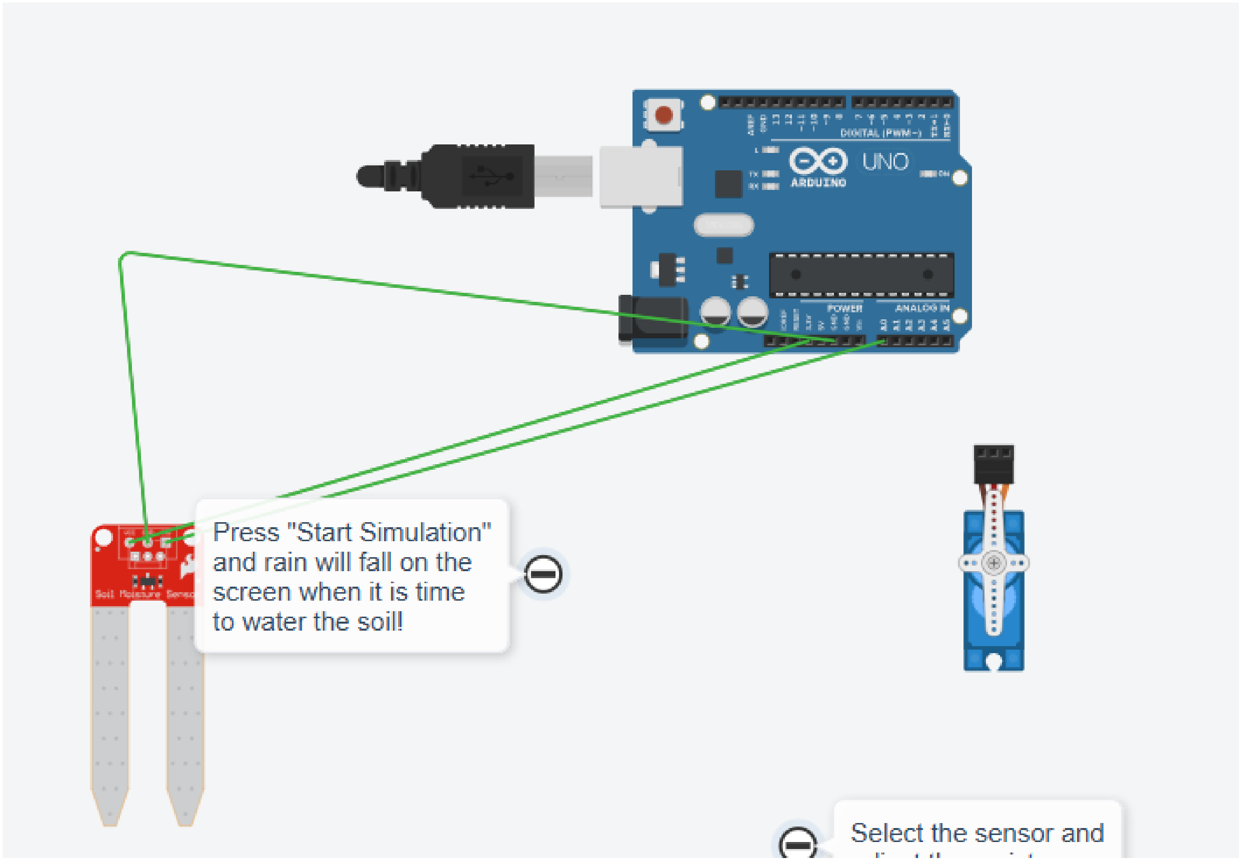

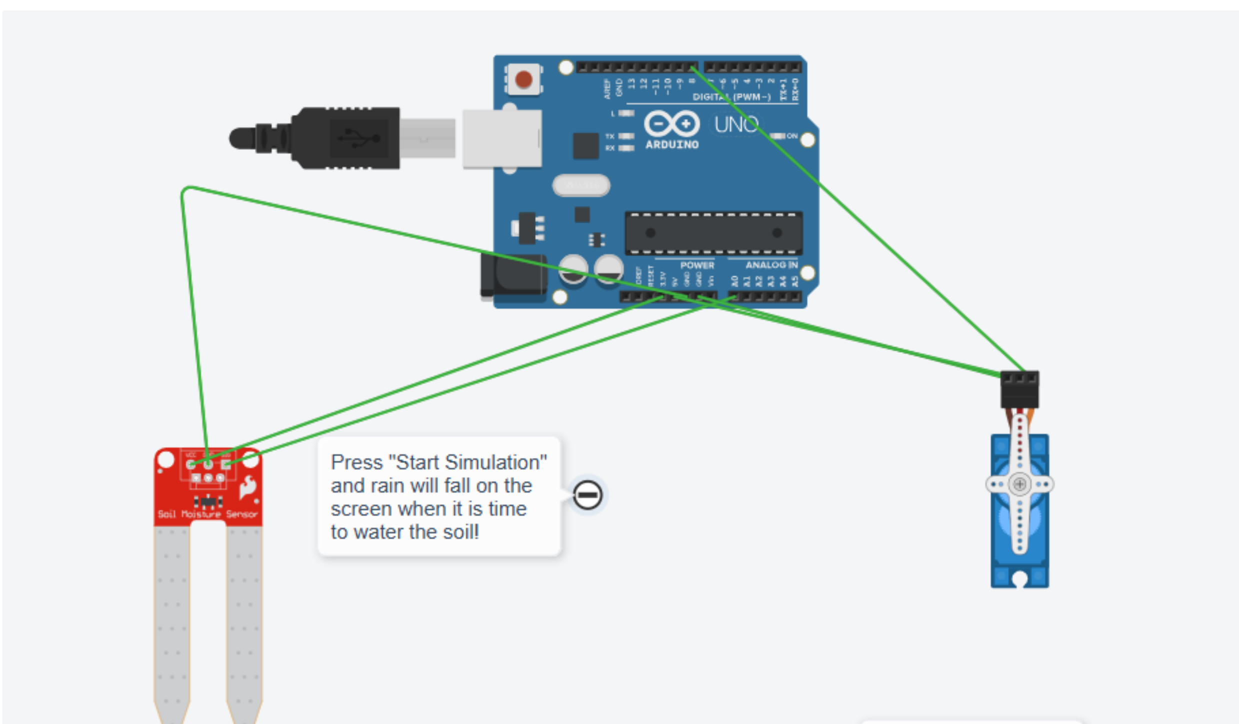

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Conclusion:- This project detects rain using a **soil moisture sensor** and triggers an **automatic shelter mechanism** using a **servo motor**. When rain is detected, the servo motor activates to cover the clothes, preventing them from getting wet.